

DATASHEET · A.1

PILLAR A

Inventory

Stock position per node, reservations and transfers.

Topology	n-node, multi-warehouse
Sync	event-sourced, deterministic merge
Reservations	per-channel holds, auto-expiry
Offline	node-local ledger, reconcile on reconnect
Latency	< 100 ms in-site, sub-second across channels

Stock	Transfers	Reservations	Transfer			
SKU	WH-A	WH-B	WH-C	Total		
SKU-1042	128	64	0	192		
SKU-2210	12	40	33	85		
SKU-3087	220	0	18	238		
SKU-4410	7	7	7	21		
SKU-5002	54	90	31	175		
● Synced 0.4 s ago				5 SKUs · 3 nodes		

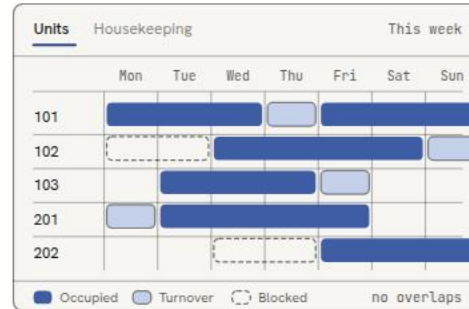
DATASHEET · A.2

PILLAR A

Units

Units, stays, availability and turnover tasks.

Allocation	overlap-free by construction
Housekeeping	status transitions create tasks
Channels	booking channels kept in step
Rates	plans, seasons and restrictions
Handover	turnover opens at check-out



DATASHEET · A.3

PILLAR A

Ledger

The transaction ledger: every entry, posted once.

Model	double-entry, append-only
Idempotency	one key per transaction
Gateways	settlement matched automatically
Close	daily close with variance report
Audit	immutable history, exportable

Sync

Close

live

POS #A-10441

24.00

18.50

VAT 4.25

Total 46.75

→

Account

Dr

Cr

1100 Cash 46.75

4000 Sales 42.50

2200 VAT 4.25

Σ Dr = Σ Cr 46.75

●

Balanced, posted once

idempotent



A.1 · Inventory

One stock position.

One stock position across every warehouse, store and channel.

[Read the specification →](#)



A.2 · Units

No two stays overlap.

Units, stays and turnover, held to a single source of availability.

[Read the specification →](#)



A.3 · Ledger

Σ debit = Σ credit.

Every sale, online or in store, posts once to one ledger.

[Read the specification →](#)

term-02 · sale.created

Stock	Transfers	Reservations	Transfer		
SKU	WH-A	WH-B	WH-C	Total	
SKU-1042	128	64	0	192	
SKU-2210	12	40	33	85	
SKU-3087	220	0	18	238	
SKU-4410	7	7	7	21	
SKU-5002	54	90	31	175	

● Synced 0.4 s ago
5 SKUs · 3 nodes

stock.reserved · SKU-1042 -1

Watch a sale move through the system.

t+000 ms A sale is created at a terminal

t+008 ms The node posts it to the ledger

t+019 ms Stock is reserved for the order

t+041 ms Every terminal shows the new state

An illustrative sequence. In the page, the figures change and the cell lights up as each step arrives.

الواجهة الحيّة: نافذة الواجهة تتحرك فعلياً: يصل حدث بيع فيتغير الرقم وتُضاء الخلية ويظهر الحدث. يُرى المنتج وهو يعمل لا صورة له، وبجانبه شرح الخطوات بالمللي ثانية (الأرقام توضيحية).

One ordinary day, three modules doing their work.

08:40

A scanner at warehouse B



Inventory

Stock for SKU-1042 updates in every channel.

14:15

A booking arrives from a channel



Units

Unit 102 is confirmed. No overlap is possible.

18:05

A sale closes at the till



Ledger

The entry posts once and the day balances.



A.1

Inventory

Stock position per node, reservations and transfers.

Topology

Sync

Reservations

Offline

[Open the specification →](#)



A.2

Units

Units, stays, availability and turnover tasks.

Allocation

Housekeeping

Channels

Rates

[Open the specification →](#)



A.3

Ledger

The transaction ledger: every entry, posted once.

Model

Idempotency

Gateways

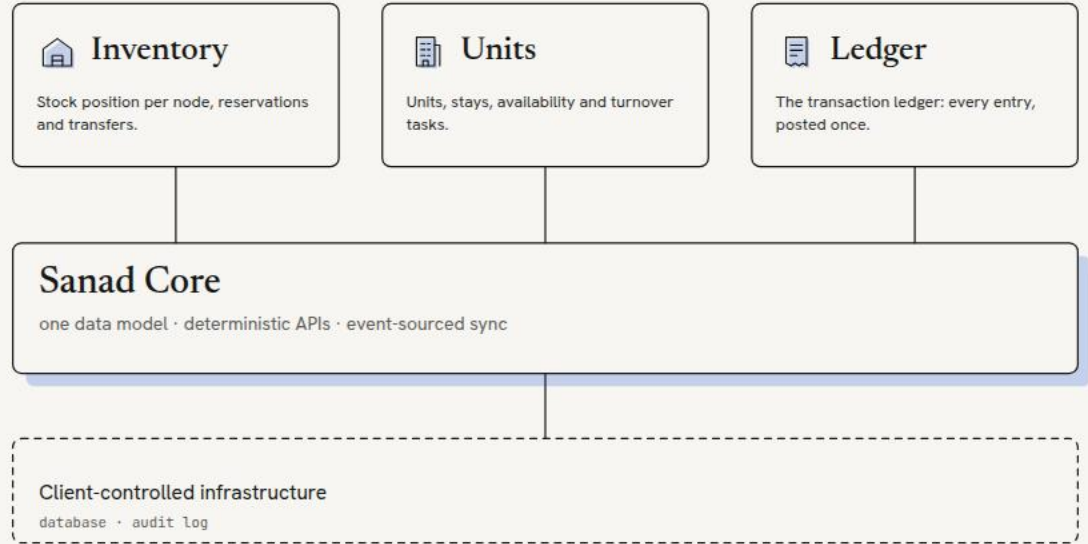
Close

[Open the specification →](#)

Three modules. One core.

Modules share one data model, so a bespoke component can replace or extend one without a rebuild.

Select a module to see its interface, schema and parameters.



Choose what you run.



Inventory

One stock position across every warehouse, store and channel.



Units

Units, stays and turnover, held to a single source of availability.



Ledger

Every sale, online or in store, posts once to one ledger.

Your configuration

Inventory and Ledger

Inventory

Topology	n-node, multi-warehouse
Sync	event-sourced, deterministic merge
Reservations	per-channel holds, auto-expiry

Ledger

Model	double-entry, append-only
Idempotency	one key per transaction
Gateways	settlement matched automatically

[Request this configuration →](#)

Opens an email with your selection filled in.

A.1
Inventory

Stock position per node, reservations and transfers.

A.2
Units

Units, stays, availability and turnover tasks.

- Allocation
- Housekeeping
- Channels
- overlap-free by construction
- status transitions create tasks
- booking channels kept in step

A.3
Ledger

The transaction ledger: every entry, posted once.

Interface wireframe - Units

Units	Housekeeping						This week
	Mon	Tue	Wed	Thu	Fri	Sat	Sun
101	Occupied			Turnover	Occupied		
102	Blocked		Occupied				Turnover
103		Occupied			Turnover		
201	Turnover	Occupied					
202			Blocked		Occupied		
Occupied Turnover Blocked							no overlaps

A.1
Inventory

keeps **one stock position** across every warehouse, store and channel, merged **event by event**, **working offline** and reconciling on reconnect.

A.2
Units

holds **one source of availability** for every stay and turnover, **overlap-free by construction**, with housekeeping tasks opened at check-out.

A.3
Ledger

posts every sale, online or in store, **exactly once**: **double-entry**, **append-only**, matched to gateway settlement.

A.1

Inventory

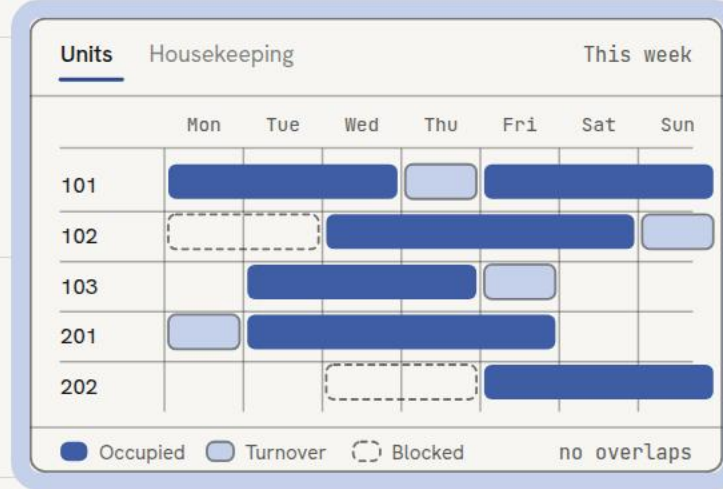
Stock position per node, reservations and transfers.

A.2

Units

A.3

Ledger



s, availability and turnover tasks.

action ledger: every entry, posted once.